

Deep Learning (AI 360): Assignment #1

Due on Tuesday, September 29, 2026 at 11:59pm PST

In this assignment you will get hands-on experience with the basic concepts, operations, and interfaces used in deep learning systems. There are many frameworks that assist with the creation, execution, and optimization of deep networks. In this assignment, we will begin with the manual implementation of networks in numpy, and then progress to using **PyTorch**. This is a popular framework with a Python interface (and another popular option is JAX). The basics learned in this assignment will give you hands-on experience with deep networks and deep learning so that you can work with layers, models, and updates and think through implementational, algorithmic, and computational issues.

Problem 1 (20 points)

Consider a deep network with N input units denoted by $\mathbf{x} \in \mathbb{R}^N$ vector, M output units denoted by $\hat{\mathbf{y}} \in \mathbb{R}^M$ vector, and K hidden units denoted by $\mathbf{h} \in \mathbb{R}^K$ vector. The hidden layer has an activation function σ (e.g., a sigmoid or ReLU). The resulting equations that govern the behavior of this simple network are:

$$\begin{aligned}\mathbf{z} &= \mathbf{W}^{(1)}\mathbf{x} + \mathbf{b}^{(1)} \\ \mathbf{h} &= \sigma(\mathbf{z}) \\ \hat{\mathbf{y}} &= \mathbf{W}^{(2)}\mathbf{h} + \mathbf{b}^{(2)}\end{aligned}$$

The loss function $\mathcal{L}$ involves a squared error on the difference between the prediction $\hat{\mathbf{y}}$ and the truth $\mathbf{y}$, $\mathcal{E}$, plus a regularizer, $\mathcal{R}$, as follows:

$$\begin{aligned}\mathcal{L} &= \mathcal{E} + \mathcal{R} \\ \mathcal{R} &= \mathbf{r}^T \mathbf{h} \\ \mathcal{E} &= \frac{1}{2} \|\mathbf{y} - \mathbf{s}\|^2\end{aligned}$$

where $\mathbf{r}$ and $\mathbf{s}$ are given (e.g., $\mathbf{s}$ is a ground truth prediction for $\mathbf{x}$).

- (a) Draw the computational graph relating $\mathbf{x}$, $\mathbf{z}$, $\mathbf{h}$, $\mathbf{y}$, $\mathcal{L}$, $\mathcal{E}$, $\mathcal{R}$. Note, in this case we expect nodes to represent collections of values, not each unit individually.
- (b) Derive the backpropagation equations for computing $\frac{\partial \mathcal{L}}{\partial \mathbf{x}}$. Use $\sigma(\mathbf{h}) = \frac{1}{1+e^{-\mathbf{h}}}$.

Credit: Problem 1 is adopted from University of Toronto's CSC421/2516 Problem Set 1.

Problem 2 (80 points)

The second half of the assignment requires Google Colab and is distributed as a notebook. All instructions are given in the notebook itself. The assignment is self-contained and imports any necessary libraries in each

section. Once the assignment is done, hand in both the written portion and the notebook portion using Canvas. When handing in programming assignment make sure you hand in all the text, code, and outputs of the executed code.

Please refer to the prerequisite document and the material for the first discussion section for a review of the basics and the colab platform. For issues with the assignment or colab, please post on piazza and attend office hours for help.